

Assignment 2: Working with Databases

Learning Objectives:

In this exercise, you will learn about various levels of the data modeling process and how that leads to schema design and implementation. More specifically, you will learn how to design and implement a simple backend database for a blogging platform using SQL / NoSQL DBMS.

Due Date: February 6, 2025

Evaluation: 10% of the total grade

Instructions

You are provided with a small dataset (blogging_dataset.csv) for this assignment. The dataset consists of users who write posts and comments on the posts. Posts and comments are timestamped with unique identifiers for the both. Comments are anonymous, i.e., the authors of comments are not revealed. You are tasked to perform the following steps using the dataset provided:

1. Analyze the dataset and create the Entity Relationship Diagram that conceptually models the data structure in the given dataset.
2. Translate the conceptual model to the physical model considering a DBMS of your choice (PostgreSQL or MongoDB). Justify your choice of the DBMS (SQL / NoSQL).
3. Implement the schema and perform the following queries:
 - a) Retrieve all users
 - b) Retrieve all posts
 - c) Retrieve all comments
 - d) Update the user email of sarah_c to sarab@gmail.com
 - e) Delete comment id cid4
 - f) What happens if you insert the data below into the user table?
User-id: uid5, Username: John_d, Email: john@go.com , Password: password123
4. If you have implemented the database using Postgres (MongoDB), which features of the database do you think would improve by switching to MongoDB (Postgres)?

Submit your report with embedded screenshots to substantiate your work.

Note: You may use either the shell or the UI for the DBMS used.

Scoring Rubric

Category	Criteria	Scores
Data modeling	All three levels of data modeling covered and explained	2
Database Implementation	correct implementation of schema, all queries and tasks performed producing correct answers	6
Report	A well-organized report containing an executive summary followed by properly labeled sections showing results and brief explanations	2